CS & IT ENGINEERING



Computer Networks

ISO OSI STACK

DPP 02 Discussion Notes



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[MCQ]

#Q. A packet has arrived with an MF(More fragments) bit value of 1. Is this the first fragment, the last fragment, or a middle fragment?

MF = 1 (atleast one)

MF = 0 (last fragment)

A. First fragment

B. Last fragment

C. Middle fragment

D. More information required

[MCQ]

#Q. In an IPV4 packet, the value of the HLEN field is 1100 in binary. How many bytes of options field is being carried by this packet?

$(1100)_2 = (12)_{10}$

A. 28 byte

B. 20 byte

C. 8 byte

D. 40 byte

Header size = 12 x 4 = 48 B

Min - 20 B

option field size = 48 - 20 = 28 B

[MCQ]

#Q. What is the maximum payload size at the Network layer and the Data link layer on the TCP/IP protocol?

Application layer

TL = 65495 B

NL = 65515 B

DL = 1500 B

X A Any size, 1480 bytes

B 65,495 bytes, 1500 bytes

C 65,535 bytes, 1500 bytes

D 65,515 bytes, 1500 bytes

[MCQ]

#Q. Consider an IP link with a maximum transfer unit of 700 bytes, and each datagram size to send is 2400 bytes. Assume that the datagram recently sent has an identification number of 422. What is the total number of fragments that must be sent to transfer the full datagram, and what is the offset value of the 2nd fragment from all the fragmented datagrams? Note: Consider the minimum header size of the IPV4 datagram.

A 4, 85

X B 5, 680

C 4, 1320

X D 1,0

2400 B = 2380 + 20

680 + 20 (700) → F_1

680 + 20 (700) → F_2

680 + 20 (700) → F_3

370 + 20 → F_4

Offset of 2nd fragment

$= \frac{680}{8} = 85$

[MSQ]

#Q. Which of the following assertions is/are TRUE about the Internet Protocol (IP)?

TTL (Time to Live)

S D

A. It is possible for a computer to have multiple IP addresses.

B. IP packets from the same source to the same destination can take different routes in the network.

C. IP ensures that a packet is discarded if it is unable to reach its destination within a given within a gi number of hops.

D. The packet source cannot set the route of an outgoing packets; the route is determined only by the routing tables in the routers on the way.

Strict routing

[MCQ]

#Q. Suppose a packet of 1800 bytes with minimum header size enters a local area network with an MTU (Maximum transfer unit) of 500 bytes. Then, how many fragments will be generated from the packet, and what is the payload size of the last packet?

A. 3, 300

B. 4, 340

C. 5, 320

D. 3, 340

Total data bytes = 1800 - 20 = 1780

Data in F_1 = 500 - 20 = 480

Data in F_2 = 500 - 20 = 480

Data in F_3 = 500 - 20 = 480

Remaining data bytes = 340

Size of last packet = 340 + 20 = 360

[MCQ]

#Q. If the value in the HLEN field of the IP header is 8, then what is the header's length?

Header length : 8 × 4 : 32 B

A 8 bytes

B 16 bytes

C 32 bytes

D 40 bytes

[MCQ]

#Q. A TCP message consisting of 2800 bytes is passed to IP for delivery across two networks. The first network can carry a maximum payload of 1500 bytes per frame, and the second network can carry a maximum payload of 500 bytes per frame, excluding network overhead. Assume that the IP overhead per packet is 20 bytes. What is the total IP overhead in the second network for this transmission?

N_1

2800 = 1500 + 1300

Fragmentation = 1496 + 1304

N_2

1496 = 496 + 496 + 496 + 8] 7 × 20 = 140 B

1304 = 496 + 496 + 312

A. 140 bytes

B. 80 bytes

C. 120 bytes

D. 160 bytes

THANK - YOU